GGG-S-66C
November 8, 1978
SUPERSEDING
Fed. Spec. GGG-S-66B
February 20, 1967
(See 6.4)

FEDERAL SPECIFICATION

SAWS, HOLE: BLADES, ARBORS AND INTEGRAL UNITS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers hole saws, their blades and arbors and those hole saws with arbors integrally welded to the saw.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 The hole saws shall be of the following types, as specified (see 6.2):

Type I - Standard (separate arbor)
Type II - Integral (permanent attached arbor)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. H28 - Screw-Thread Standards for Federal Services.
Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are availble without charge from General Services Administration Business Service Centers in Boston; New York; Philadelphia; Washington, DC; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications, Standards, Drawings, and Publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents, form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Iron and Steel Institute (AISI):

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 1000 - 16th Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

3. REQUIREMENTS

3.1 Qualification. The hole saws and arbors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 General requirements.

3.2.1 Illustrations. The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of hole saws or arbors otherwise in accordance with this specification.

3.2.2 Interchangeability. Hole saws and arbors from the same manufacturer shall be manufactured to standards which will permit replacement or adjustment without modification and will insure ease of interchangeability.

3.2.3 Construction. The hole saws and arbors shall be constructed of parts which are new, without defects, and free of any characteristics that will prevent them from passing any of the examinations or tests specified herein.

3.2.4 Screw threads. All threaded parts of the hole saws and arbors shall conform to Fed. Std. No. H28.

3.2.5 Finish. The hole saws and arbors shall be clean, free from burrs, excessive welding or brazing material, and shall be provided with a protective coating to preclude corrosion.

3.2.6 Identification marking. Each hole saw and arbor shall be marked in a plain and durable manner with the size and manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.2.7 Workmanship. Workmanship of the hole saws and arbors shall be such that they shall be free from defects which would adversely affect the operation efficiency, serviceability, and life expectancy of them.

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3.3 Detail requirements.

3.3.1 Design. Types I and II hole saws shall be designed to saw holes through machinable materials up to and including 1-1/8 inch thick. The hole saw shall consist of a single solid piece cutting edge and back of high speed steel, or a high speed steel cutting edge welded continuously to an alloy steel back, or may be constructed from high speed steel flat stock. In forming the body of the saw, the butt seam shall be welded or brazed. Saw teeth adjacent to the butt seam shall not be annealed by the fabrication. Saws 5/8 inch through 1-3/16 inch diameter may be manufactured from high speed steel round stock. Unless otherwise specified (see 6.2), the cutting edge shall have six teeth per inch measured on the circumference, sharpened and set to provide a free cutting edge. Holes, slots, or other suitable means shall be provided to facilitate removal of slugs. The blades shall be heat treated and tempered to withstand the tests of section 4. The thickness for the hole saw blade and body shall be not less than 0.040 inch for sizes 1-5/16 inch in diameter and smaller and 0.047 inch for saws sized larger than 1-3/8 inch diameter. Hole saws shall be furnished with or without arbors or integrals, as specified (see 6.2).

3.3.2 Hole saw size. Types I and II hole saws shall be of the sizes specified in table I.

TABLE I. Hole saw sizes.

Diameter (inch)	Pipe tap size (inch)	Pipe entrance size (inch)	Diameter (inch)	Pipe tap size (inch)	Pipe entrance size (inch)
5/8			2-5/16		
11/16			2-3/8		
3/4		3/8	2-1/2		2
13/16			2-9/16		
7/8	3/4	1/2	2-5/8	2-1/2	
15/16			2-3/4		
1			2-7/8		
1-1/16			3		2-1/2
1-1/8	1	3/4	3-1/8		
1-3/16			3-1/4	3	
1-1/4			3-3/8		
1-5/16			3-1/2		
1-3/8		1	3-5/8		3
1-7/16			3-3/4	3-1/2	
1-1/2	1-1/4		3-7/8		
1-9/16			4		
1-5/8			4-1/8		3-1/2
1-11/16			4-1/4		
1-3/4	1-1/2	1-1/4	4-3/8		
1-13/16			4-1/2		4
1-7/8			4-3/4	4-1/2	
2		1-1/2	5		
2-1/16			5-1/2		
2-1/8			5-3/4		
2-1/4	2		6		

3.3.3 Hole saw concentricity. The total indicator runout (T.I.R.) of the hole saw shall not exceed the tolerance shown in table II.

TABLE II. Tolerance for tooth crest runout and concentricity.

Saw diameter (inches)	Concentricity of saw body below gullet of teeth (inch)	Maximum tooth crest runout (inch)	
		At brazed seam T.I.R.	Around periphery T.I.R.
5/8 through 1-7/8	0.020	0.010	0.030
2 through 3-7/8	.030	.010	.040
4 through 6	.040	.010	.060

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3.3.4 Arbor drive capacity. Types I and II hole saw arbors shall be capable of driving the size hole saw specified in table III.

TABLE III. Drive capacity.

Hole saw size (inch)	Arbor numbers	
	Type I saw	Type II saw
5/8 - 1-3/16	0, 1	7/16 Hex
1-1/4 - 2-7/8	2, 3	7/16 Hex
3 - 4-1/8	2, 3, 4	7/16 Hex
4-1/4 - 6	3, 4	7/16 Hex

3.3.5 Arbor dimensions. Arbor dimensions for types I and II hole saws shall conform to the dimensions specified in table IV.

TABLE IV. Hole saw arbor dimensions.

	Arbor designation	Shank size diameter or width across flats as applicable A plus 0.005	Body diameter B plus 5/32 minus 1/8	Hole saw thread size C class 2A thread	Drive pin size D plus 0.000 minus 0.020	Pilot drill size E nominal diameter
	(Number)	(Inch)	(Inches)		(Inch)	(Inch)
Type I	0	1/ 1/4 dia. or Hex	3/4 dia. or Hex	1/2-20		1/4
	1	7/16 Hex	3/4	1/2-20		1/4
	2	7/16 Hex	1-1/2	5/8-18	0.250	1/4
	3	5/8 Hex	1-1/2	5/8-18	0.250	1/4
	4	No. 3 Morse taper	3-15/16		5/16-18 thds.	3/8 Pilot pin
Type II	7/16 Hex	0.430 ±0.002	3/4 +0.0125 -0.000 / 1 +0.0125 -0.0312	None	None	1/4

1/ Pilot drill may extend through arbor body and act as shank.

3.3.6 Pilot drill. Types I and II hole saws shall be furnished with a 1/4 inch (letter size E) high speed steel drill. The hole in the arbor for the pilot drill shall be concentric with the drill shank when attached within 0.008 inch T.I.R. The pilot drill shall be similar to figure 8, contain a notch or flat which will locate the point of the drill approximately 1/4 inch below the cutting edge of the saw and shall provide positive drive between the arbor and the drill. The drill shall have a minimum flute length of 1-13/16 inch being located in the arbor so that the full length of the flute shall protrude beyond the cutting teeth allowing the shank of the drill to center and guide the saw when used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Qualification inspection. Qualification inspection shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in 4.5.

4.3.1 Samples for qualification testing. Six sample hole saws (with corresponding arbors) for each of the representative sizes shown in table V shall be submitted for qualification testing of the hole saw.

TABLE V. Representative samples and qualification test requirements.

Hole saw size diameter (inches)	Driving arbor	Cutting speed (r/min)	Applied load (pounds) (maximum)	Number of holes to be completed
0.625	No. 1	367 to 400	100	20
2	No. 3	115 to 160	215	6
4	No. 3 or 4	58 to 80	436	3
6	No. 3 or 4	38 to 50	664	2

4.4 Quality conformance inspection.

4.4.1 Lot. A lot shall consist of all hole saws or arbors of one size offered for inspection at one time. The sample unit shall be one completely fabricated hole saw or arbor.

4.4.2 Sampling for visual examination. A random sample of hole saws and arbors shall be selected from each inspection lot offered for visual examination in accordance with MIL-STD-105, inspection level II with an acceptable quality level (AQL) of 2.5 percent defective.

4.4.3 Sampling for dimensional examination. A random sample of hole saws and arbors shall be selected from each lot for dimensional examination in accordance with MIL-STD-105, inspection level S-4 with an AQL of 2.5 percent defective.

4.4.4 Visual examination. Each sample unit selected in accordance with 4.4.2 shall be visually examined to verify compliance with this specification. Examination shall be conducted as specified in table VI. Any hole saw or arbor in the sample containing one or more visual defect(s) shall not be offered for delivery. If the number of defective hole saws or arbors in the sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

TABLE VI. Visual examination of defects.

Examine	Defects
Design	Size of hole saw or arbor not as specified Minimum depth of cut of hole saw not as specified Drive pins missing on arbors numbers 2, 3 Pilot drill missing Pilot drill flute length not as specified
Material	Teeth not of high speed steel Pilot drill not high speed steel
Construction	High speed steel cutting edge not welded continuously Arbor not as specified Hole saw or arbor with parts missing, defective, cracked, split, or evidence of rust Notch missing from pilot drill Teeth not properly set and sharpened Drive holes in hole saw missing or not properly located Threads of hole saw and arbor wrong size, impaired, or missing
Finish	Not as specified
Marking identification	Missing, incomplete, not legible, not as specified

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4.4.5 Dimensional examination. Each sample unit selected in accordance with 4.4.3 shall be dimensionally examined to determine compliance with the dimensional requirements of this specification. Any dimension not within specified tolerance limits shall be classified a defect. Any hole saw or arbor in the sample containing one or more dimensional defect shall not be offered for delivery. If the number of defective hole saws or arbors in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.5 Qualification tests.

4.5.1 Determination of concentricity. Three sample hole saws (with arbors) of each representative size specified in table V, shall be tested for radial and tooth crest runout by means of dial surface gages and a holding fixture similar to that shown on figure 9. If one of the three samples of a representative size fails to conform to the required tolerance, three additional samples for this group shall be selected for measurement. If one of the three additional samples fails to conform, this shall be cause for rejection.

4.5.2 Cutting test. Three hole saws for each representative size, which have passed the tests for determination of concentricity, shall be subjected to a cutting test, applicable to the conditions shown in table V. The test log material shall be 3/8 inch thick A.I.S.I. type 304 stainless steel in the hot rolled annealed and pickled condition. The saw shall be positioned over the test log in such a manner to permit an overlap of approximately 1/16 inch over one edge. This overlap shall provide a method for removing the chips from the kerf. A suitable commercial grade of cutting oil shall be used as a lubricant. The method of applying load may be dead load, cantilever, or hydraulic. If more than one saw of any representative size fails to complete the number of holes specified in table V, this shall be cause for rejection. Failure shall be considered to have occurred when the saw breaks or stops cutting.

4.6 Examination of preparation for delivery. An examination shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored in accordance with table VII. The sample unit shall be one shipping container fully prepared for delivery, selected just prior to the closing operation. Defects of closure listed in table VII shall be examined on shipping containers fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an AQL of 4.0 percent defective.

TABLE VII. Examination of preparation for delivery defects.

Examine	Defect
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Any nonconforming component; component missing, damaged or otherwise defective affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, inadequate nailing and stapling, loose strapping, distortion of containers.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. All surfaces of the saw shall be cleaned, thoroughly dried, and preserved with a soft film corrosion-preventive compound. The preservative shall be either a hot application petrolatum compound or cold application solvent cutback compound. Each coated saw shall be wrapped with aluminum foil or heavy duty flexible, greaseproofed, waterproofed, noncorrosive barrier material. The foil or barrier shall be loosely applied to the saw and secured in place by taping, tying, or other suitable means. Projections and sharp edges of the saw which may damage the barrier shall be cushioned. Each wrapped saw shall be packaged in a close-fitting paperboard box. The paperboard shall be highly sized to afford water resistance and shall be a folding or set up grade. The boxes shall be constructed with metal stays or lap joints that are effected with waterproof adhesive. The style of the boxes shall be at the option of the contractor. The boxes shall be closed and sealed with waterproof pressure-sensitive adhesive packaging tape.

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5.1.2 Commercial. Each saw shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the item against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

5.2.1 Level A. Saws, preserved and packaged as specified in 5.1, shall be packed in nailed wood, cleated plywood, paper overlaid veneer or wirebound wood boxes conforming to Uniform Freight Classification Ratings, Rules, and Regulations or National Motor Freight Traffic Association Classes and Rules. The construction of the boxes shall be such that they are suitable for overseas shipment and outdoor, open storage. The boxes shall be closed and strapped with flat steel strapping measuring not less than 1/2 x 0.020 inch. When the gross weight exceeds 200 pounds, each box shall be constructed with skids or a skid base to provide for access by forklift.

5.2.2 Commercial. The packaged saws shall be packed in fiberboard boxes to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under Uniform Freight Classification Ratings, Rules, and Regulations or National Motor Freight Traffic Association Classes and Rules.

5.3 Marking. Marking shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. The hole saws covered by this specification are intended to be used for sawing round holes in metal, plastic, wood, and other similar materials.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of hole saw required (see 1.2.1).
(c) Number of teeth per inch, if different (see 3.3.1).
(d) Whether hole saws are to be furnished with/without arbors or integrals (see 3.3.1).
(e) Size of hole saw (see 3.3.2).
(f) Size of arbor (see 3.3.4).
(g) Level of preservation, packaging, and packing required (see 5.1 and 5.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List QPL-GGG-S-66 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Naval Ship Engineering Center, Department of the Navy, Washington, DC 20362, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Changes from previous issue. The symbol "#" is not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

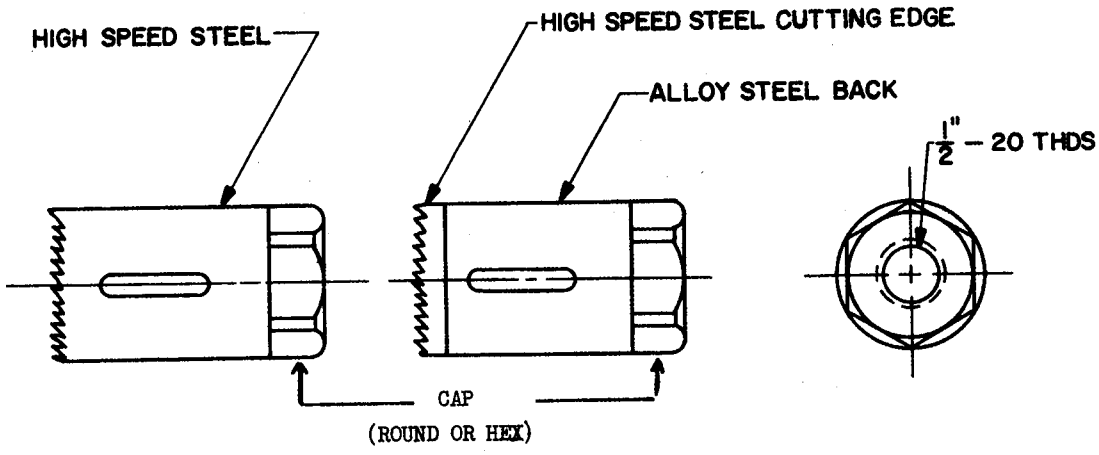
Review activity:
Army - MI

User activities:
Army - ME, SM
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Preparing activity:
Navy - SH
(Project 3455-0051)

Civil Agency Coordinating Activity:
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SH 8032

FIGURE 1. Hole saw, 1-3/16 inch diameter and smaller (without drive holes) (use arbor numbers 0 and 1).

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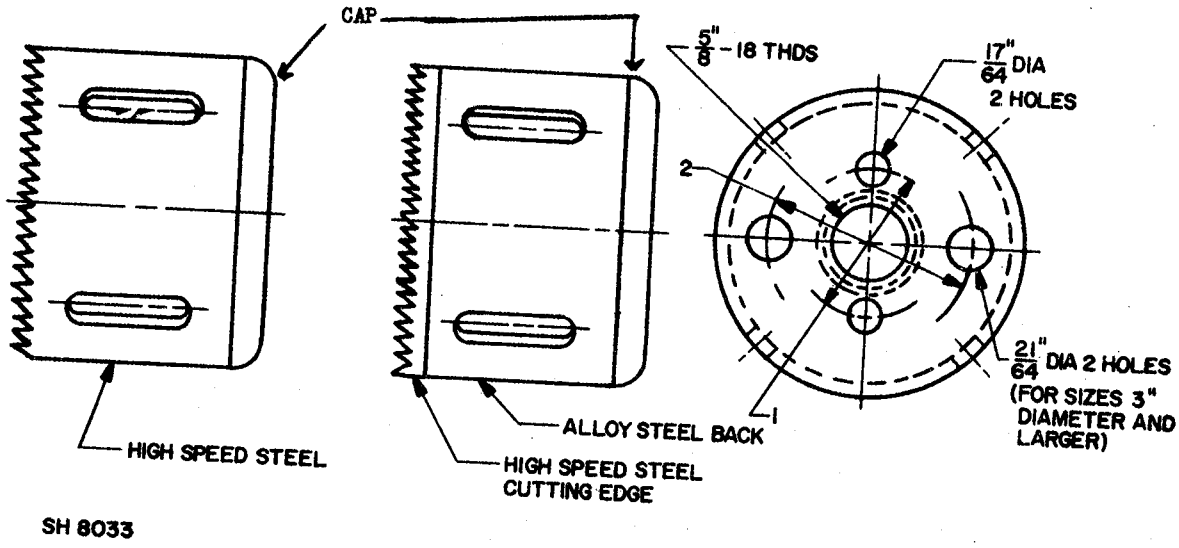
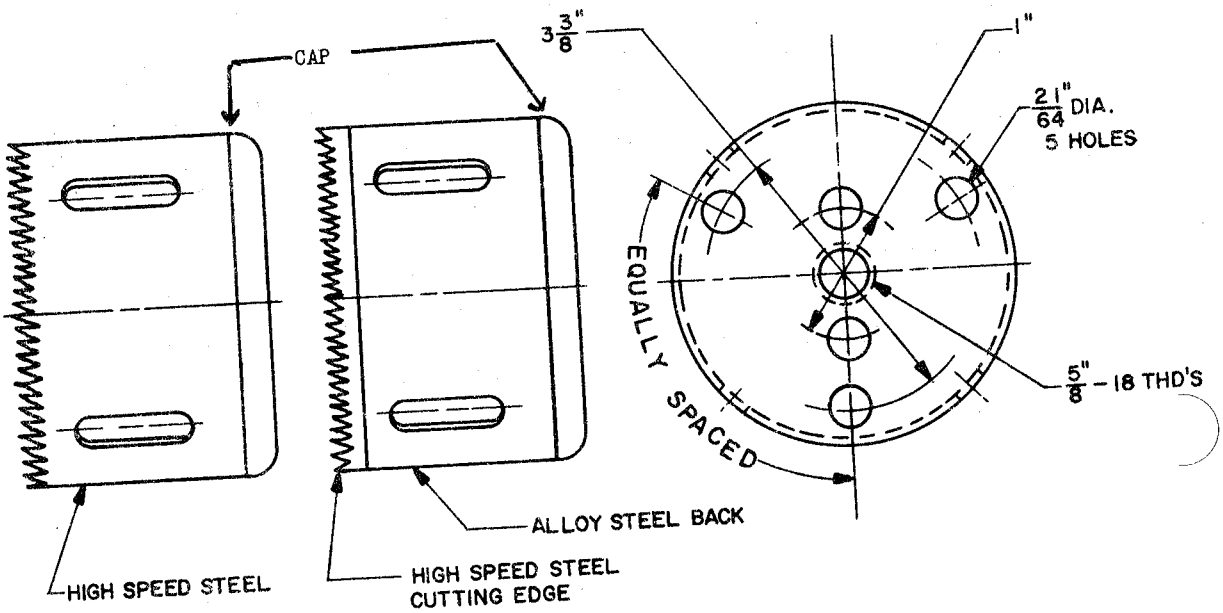


FIGURE 2. Hole saw, 1-1/4 to 4-1/8 inch diameter (with drive holes) (use arbor numbers 2, 3, and 4).

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SH 8034

FIGURE 3. Hole saw, 4-1/4 to 6 inch diameter (use arbor numbers 2, 3, and 4).

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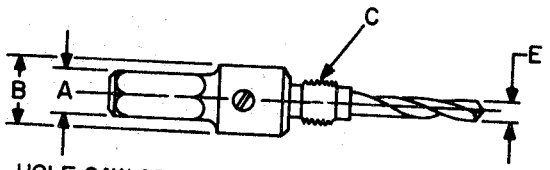


FIGURE 4. HOLE SAW ARBOR NUMBERS O AND I (WITHOUT DRIVE PINS).

SH 11298

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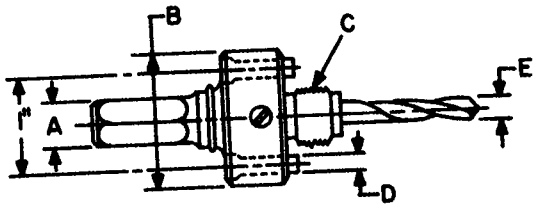


FIGURE 5. HOLE SAW ARBOR NUMBERS 2 AND 3 (WITH DRIVE PINS).

SH11299

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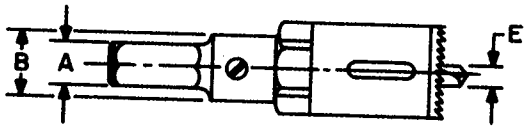


FIGURE 6. Type II hole saw.

SH11300

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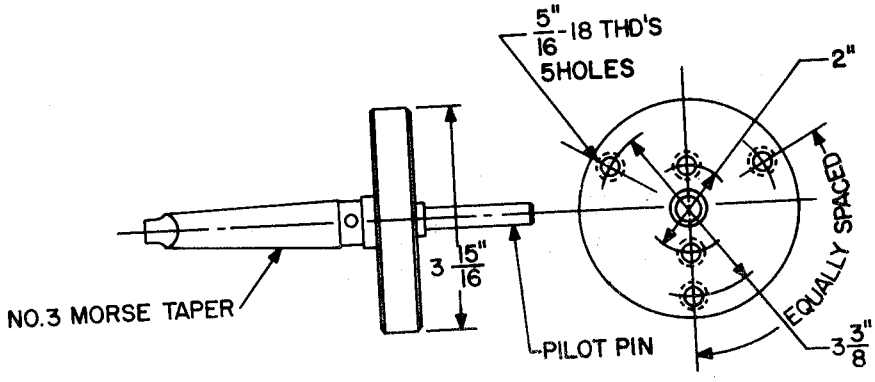


FIGURE 7. Hole saw arbor number 4 (number 3 Morse taper).

SH11301

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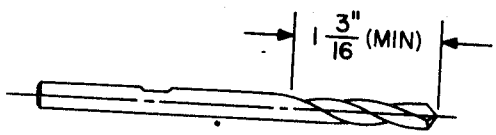


FIGURE 8. Pilot drill (modified high speed twist drill).

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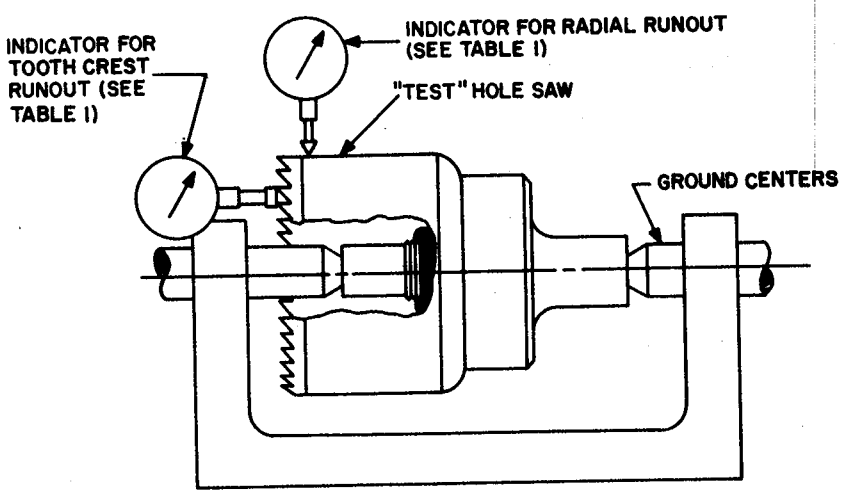


FIGURE 9. Jig for measuring tooth crest and radial runout.

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OMB Approval
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A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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